

NiveshRakshak: AI Investor Rights & Grievance Assistant

Project structure report with flow charts and worked examples

1. Executive Summary

NiveshRakshak is an AI-powered investor protection system. It does not just explain investor rights. It converts complex financial rules into **actionable guidance**, helps investors **identify issues**, **prepare grievances**, and **navigate the correct redressal process**.

Pitch line: *"From confusion to complaint to resolution, in one platform."*

Layer	What it delivers
Awareness	Rights library, quizzes, Investor Safety Score
Prevention	Scam red-flag checker (risk education)
Understanding	Document analysis ("Know Your Rights")
Resolution	Complaint classification, draft, routing, tracking

2. Problem Statement

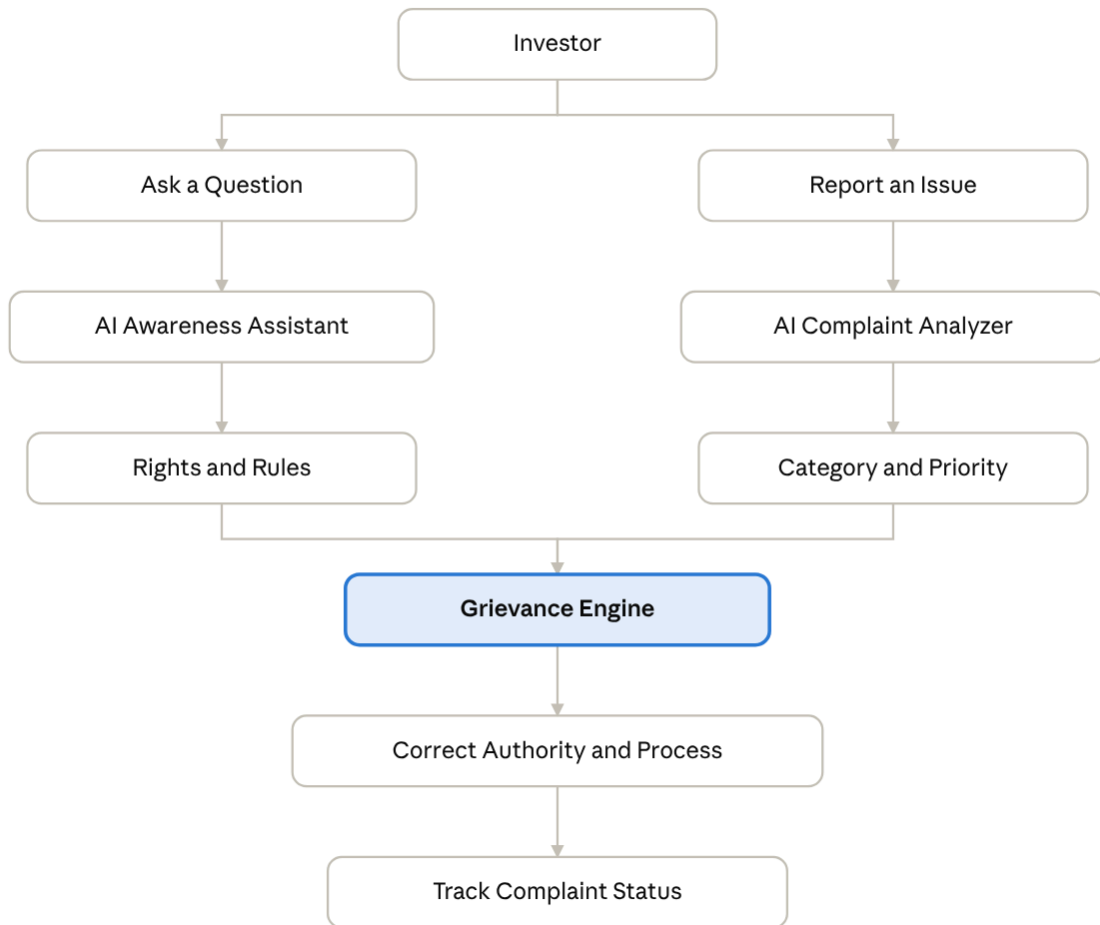
New investors commonly:

- Do not know their rights
- Cannot tell whether a broker, AMC or company is breaking rules
- Do not know where or how to complain
- Struggle with financial and legal language
- Fall into fraudulent investment schemes
- Do not know the grievance escalation process

3. High-Level System Flow

The investor enters through one of two doors: **ask a question** or **report an issue**. Both converge in the Grievance Engine.

Both paths converge in one Grievance Engine

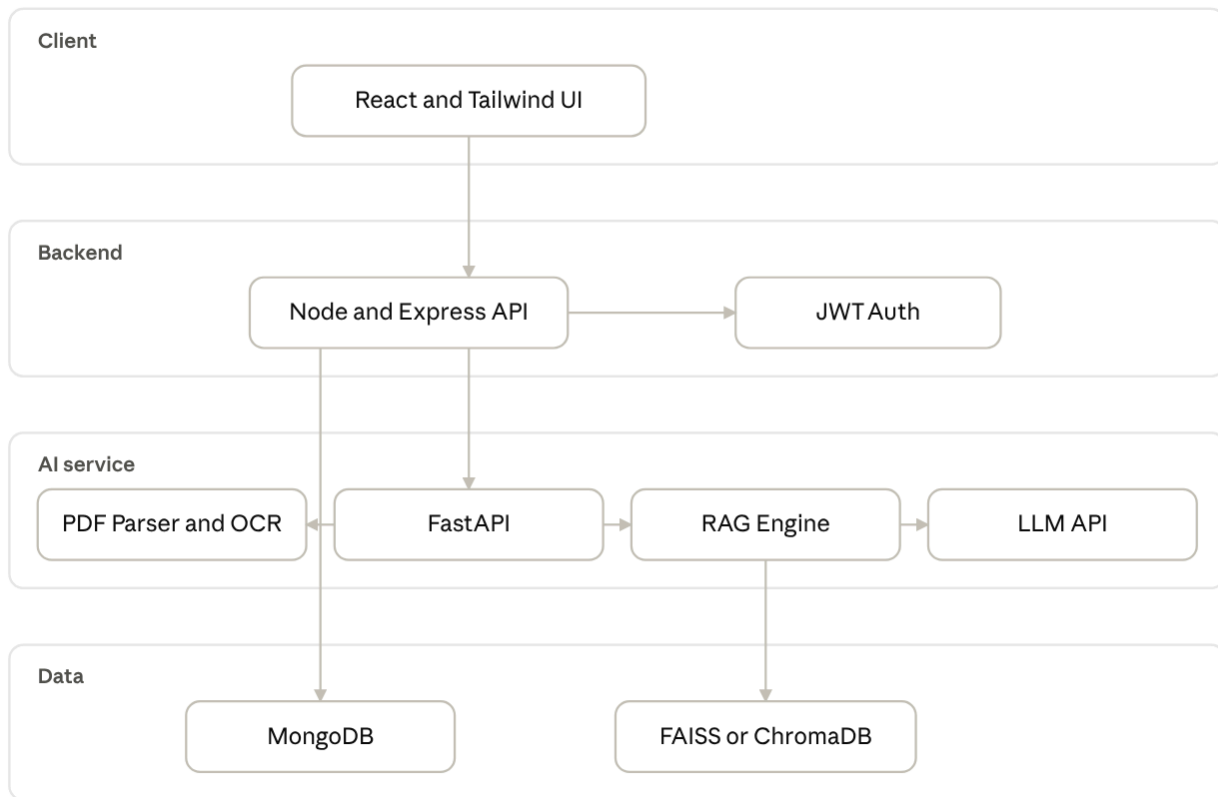


system flow · 2 entry paths, 1 engine

4. System Architecture

Four layers: a React front end, a Node API, a Python AI service, and data stores.

The Node API sits between the UI, the AI service and the data



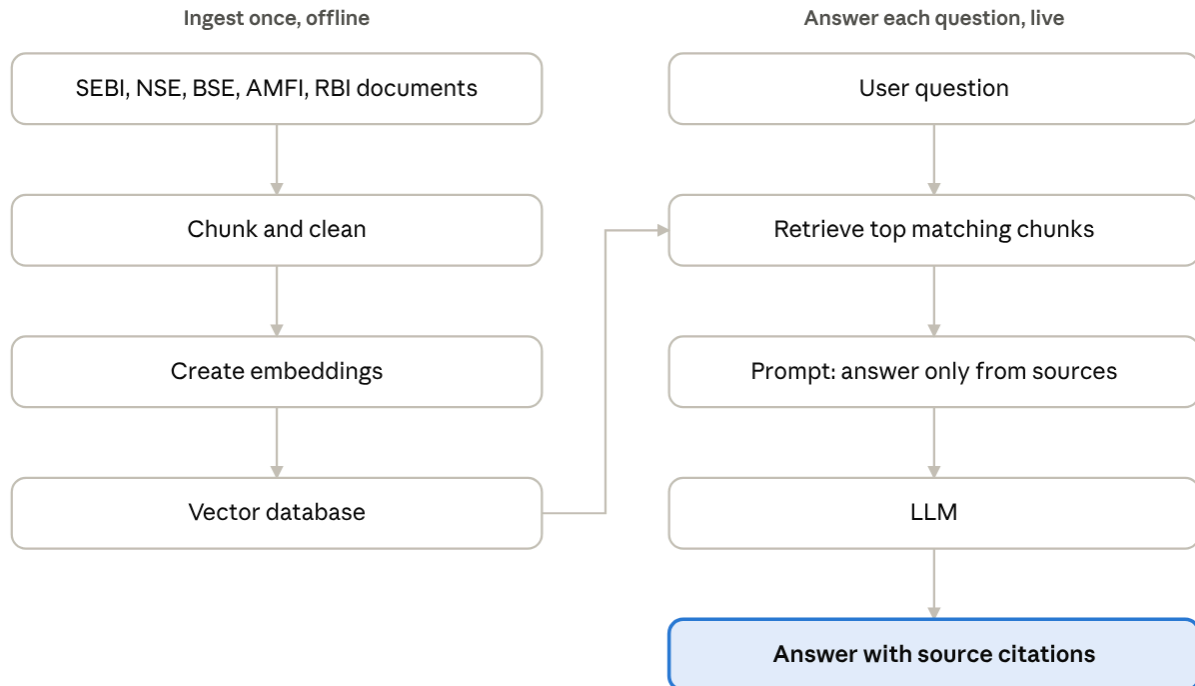
architecture · 4 layers

Layer	Technology	Responsibility
Frontend	React, Tailwind, Framer Motion, Recharts	Chat, forms, dashboard, score charts
Backend	Node.js, Express, JWT, bcrypt	Users, complaints, status, PDF export
AI service	Python, FastAPI, LLM API	Q&A, classification, drafting, scam check
Retrieval	FAISS or ChromaDB	Stores embeddings of official documents
Database	MongoDB	Users, complaints, quiz scores, documents
Documents	PDF parser, OCR	Reads statements and agreements
Deployment	Vercel (frontend), Render (backend, AI)	Hosting

5. RAG Pipeline (Why Answers Are Trustworthy)

The LLM never answers financial or legal questions from memory alone. It answers only from retrieved official text.

Answers are generated only from retrieved official text



RAG pipeline · ingest and answer lanes

Design rules

- Every answer shows its source document.
- If nothing relevant is retrieved, the system says "I could not find this" instead of guessing.
- A standing disclaimer: guidance is informational, not legal advice.

6. Feature Modules with Examples

6.1 AI Investor Rights Assistant

Input: "My broker charged me hidden charges. What are my rights?"

Output field	Example
Investor right	Right to transparent information on charges
Relevant regulation	Retrieved rule with source link

Output field	Example
Evidence to collect	Contract note, ledger, fee schedule
Action	Ask broker for a written breakup of charges
Where to complain	Broker grievance cell, then the regulator's complaint channel

6.2 Grievance Registration and Analyzer

Input

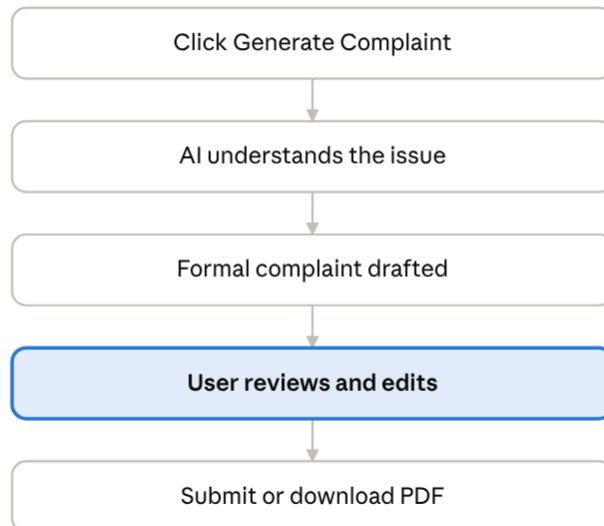
- Problem: Broker deducted ₹2,500 without explanation
- Entity: XYZ Securities
- Date: 12 Sept 2026
- Evidence: Statement and screenshots

AI-generated analysis

Field	Value
Category	Unauthorized charges
Severity	High
Evidence required	Transaction statement
Recommended action	Raise grievance with the broker first
Escalation	Appropriate regulatory grievance channel

6.3 Complaint Draft Generator

The user reviews every draft before it is submitted



complaint drafting · 5 steps

Human in the loop: the user always reviews before anything is submitted.

6.4 Scam and Fraud Awareness

Input: "Guaranteed 25% monthly return. Send ₹50,000 today."

Red flag	Detected
Guaranteed returns	Yes
Urgent payment request	Yes
Unverified entity	Yes
Unrealistic return	Yes

Risk level: HIGH

Important: present this as **risk education**, not guaranteed fraud detection.

6.5 Investor Rights Library

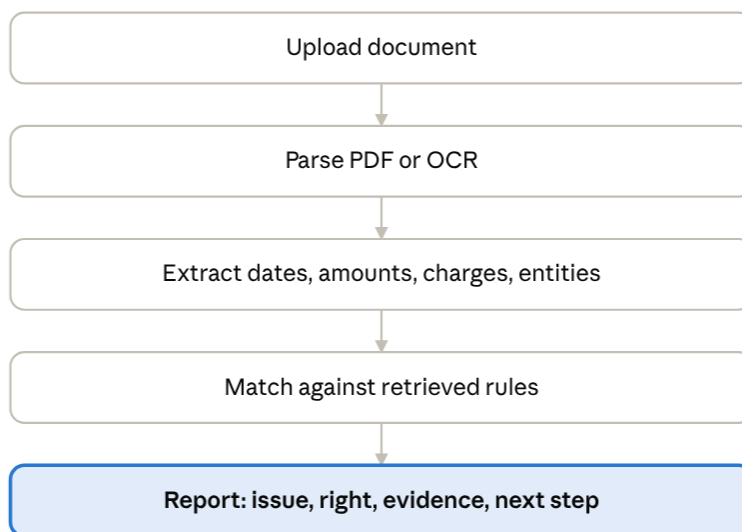
Each card follows one pattern: **Legal language** → **Simple language** → **Example** → **What to do**.

- Right to Information

- Right to Fair Treatment
- Right to Grievance Redressal
- Right to Transparency
- Right to Receive Statements
- Right to Privacy and Security

6.6 Know Your Rights (Document Analysis)

An uploaded document becomes issue, right, evidence and next step



document analysis · 5 steps

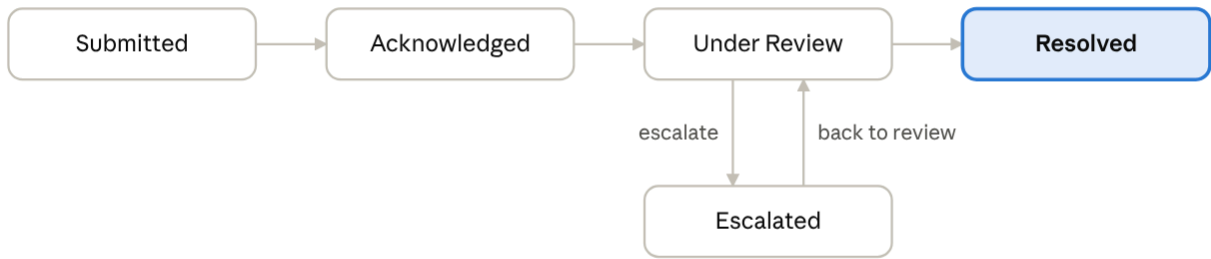
Sample result

- Potential issue: Unauthorized fee detected
- Investor right: Right to transparent information
- Evidence: Transaction dated 14/09/2026
- Next step: Raise grievance with the concerned intermediary

Accepted uploads: broker agreement, transaction statement, mutual fund statement, emails, complaint responses.

7. Grievance Lifecycle

A complaint can loop through escalation before it is resolved



The dashboard lists pending complaints and escalation deadlines.

grievance lifecycle · 5 states

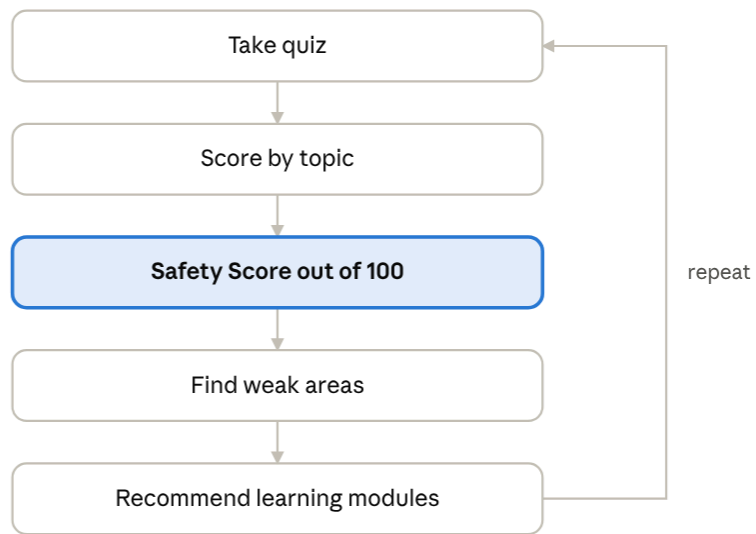
Tracker view for complaint #INV10234

Stage	Status
Submitted	✓
Acknowledged	✓
Under review	In progress
Resolved	Pending
Escalated	Pending

The dashboard lists pending complaints and escalation deadlines.

8. Investor Safety Score

Short quizzes produce a score and a personal learning path.

Weak areas feed the next round of learning

safety score loop · 5 steps

Example: 82 / 100

Area	Result
Basic investor rights	Strong
Grievance process	Strong
Scam detection	Needs work
KYC awareness	Needs work

9. Data Model (MongoDB)

Collection	Key fields	Relations
User	name, email, passwordHash	Files complaints, takes quizzes, uploads documents
Complaint	complaintId, entity, category, severity, draftText	Has status events, attaches documents
StatusEvent	stage, timestamp	Belongs to one complaint
Document	fileName, extractedText	Belongs to a user, optionally to a complaint
QuizResult	score, weakAreas	Belongs to a user

10. Suggested Folder Structure

```

niveshrakshak/
├── client/                                # React + Tailwind
│   ├── src/
│   │   ├── pages/                        # Home, Assistant, Complaint, Tracker, Library,
│   │   │   └── Score
│   │   ├── components/                  # ChatBox, RiskBadge, StatusTimeline, ScoreGauge
│   │   └── services/                    # API calls
│   └── server/                          # Node + Express
│       ├── routes/                      # auth, complaints, documents, quiz
│       ├── controllers/
│       ├── models/                      # User, Complaint, Document, QuizResult
│       └── middleware/                  # JWT auth, validation
├── ai-service/                          # Python + FastAPI
│   ├── rag/                            # ingest.py, retriever.py, prompts.py
│   ├── analyzers/                      # complaint.py, scam.py, document.py
│   └── main.py
└── data/
    └── regulatory_docs/                 # SEBI, NSE, BSE, AMFI, RBI source files

```

11. API Overview

Endpoint	Purpose
POST /api/auth/register, /login	Account and JWT

Endpoint	Purpose
POST /api/ask	Rights question to RAG answer
POST /api/complaints	Create complaint and run analyzer
POST /api/complaints/:id/draft	Generate formal complaint
GET /api/complaints/:id	Status and timeline
POST /api/scam-check	Red-flag analysis
POST /api/documents/analyze	Upload and analyze document
GET, POST /api/quiz	Questions, submit answers, score

12. End-to-End Example Journey

1. The investor uploads a statement showing a ₹2,500 deduction.
2. The server passes the file to the AI service, which parses and analyzes it.
3. The AI service finds an unauthorized fee and retrieves the matching rule.
4. The web app shows the issue, right, evidence and next step.
5. The investor clicks Generate Complaint and the AI drafts a formal complaint.
6. The investor reviews and submits; the server saves it as ID INV10234.
7. The investor receives status tracking and escalation deadlines.

13. Responsible AI and Limits

- Answers come only from retrieved official sources, with citations.
- Scam checker is **risk education**, not a fraud verdict.
- Complaint drafts require user review before submission.
- Output is guidance, not legal or financial advice.
- Uploaded documents hold sensitive data: encrypt storage, allow deletion, and never log raw content.
- Verify current grievance channels and timelines on official sites before launch, since they change.

14. Build Plan

Phase	Scope
1. Core	Auth, rights library, RAG Q&A
2. Grievance	Complaint form, classifier, draft, PDF export
3. Tracking	Status timeline and dashboard
4. Differentiators	Scam checker, document analysis
5. Engagement	Quiz and Investor Safety Score

Hackathon tip: if time is short, demo phases 1, 2 and 4. The upload-to-complaint flow in section 12 is the strongest live demo.

15. Hackathon USP

Do not pitch it as a website explaining investor rights. Pitch it as an AI-powered investor protection system covering **awareness, prevention, grievance and resolution** in one flow.